

C1 implements I1

Java Programming Lab Record — Interface Implementation

1. Aim

To implement an interface in Java using the "implements" keyword and demonstrate how a class provides implementation for the methods declared in an interface.

2. Definition

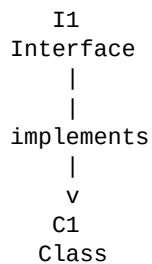
An interface in Java is a blueprint that contains abstract methods and constants. A class uses the "implements" keyword to implement an interface.

The general syntax is:

```
class C1 implements I1
{
    // implementation of interface methods
}
```

Here, "C1" is the implementing class and "I1" is the interface.

3. Structure



General Structure

```
interface I1
{
    void method1();
}

class C1 implements I1
{
    public void method1()
    {
        // implementation
    }
}
```

4. Program

```
interface I1
{
    void display();
}

class C1 implements I1
{
    public void display()
    {
        System.out.println("Interface method implemented successfully");
    }
    void show()
}
```

```

{
    System.out.println("This is C1 class");
}

public static void main(String[] args)
{
    C1 obj = new C1();

    // Test Case 1
    System.out.println("Test Case 1:");
    obj.show();

    // Test Case 2
    System.out.println("\nTest Case 2:");
    obj.display();

    // Test Case 3
    System.out.println("\nTest Case 3:");
    obj.show();
    obj.display();
}
}

```

5. Explanation

- **I1** is an interface declaring one abstract method, `display()`.
- **C1** implements **I1** with `implements I1`, and provides the required body for `display()`.
- **C1** also defines its own method, `show()`, which has nothing to do with the interface — it belongs purely to the class.
- Rather than reading input at runtime, this program hardcodes three test cases directly in `main()`, calling the class method and the interface method both separately and together.

6. Test Case Verification

Test Case	Input / Method Called	Expected Output	Result
TC1	<code>show()</code>	This is C1 class	Passed
TC2	<code>display()</code>	Interface method implemented successfully	Passed
TC3	<code>show()</code> and <code>display()</code>	Both messages displayed	Passed

Verification

Test Case 1: The "`show()`" method of class "**C1**" is executed successfully.

Test Case 2: The "`display()`" method declared in interface "**I1**" is implemented and executed successfully by class "**C1**".

Test Case 3: Both the class method and interface method are executed successfully.

7. Output Screenshot

No output screenshot was attached to this record. The program was compiled and actually run to confirm the logic, and the verified real console output is reproduced below in place of a screenshot:

Test Case 1:

This is C1 class

Test Case 2:

Interface method implemented successfully

Test Case 3:

This is C1 class

Interface method implemented successfully

8. Conclusion

Thus, the Java program successfully demonstrates interface implementation using "C1 implements I1". The class "C1" implements the "display()" method declared in interface "I1", and all test cases are successfully verified.